



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Subject: Programming with Java

ASSIGNMENT No : 4

TITLE : Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required

CONCLUSION-

This assignment demonstrated the use of **Java wrapper classes** for converting primitive data types to objects and vice versa. It provided practical experience with **autoboxing, unboxing, and parsing** methods to handle different types of data efficiently. The Student Marks program showed how string inputs can be converted into numeric values for calculations, while the Employee Payroll System illustrated the use of wrapper objects for input validation and salary computation. Overall, the assignment enhanced the understanding of wrapper classes and their importance in data conversion, validation, and real-world Java

applications.

EXERCISES-

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 1 X
J AS4EX1.java > StudentMarks > main(String[] args)
1 public class StudentMarks {
  Run main | Debug main
2   public static void main(String[] args) {
3
4       String mark1 = "85";
5       String mark2 = "90";
6       String mark3 = "78";
7
8       int m1 = Integer.parseInt(mark1);
9       int m2 = Integer.parseInt(mark2);
10      int m3 = Integer.parseInt(mark3);
11
12      int total = m1 + m2 + m3;
13
14      System.out.println("Marks 1: " + m1);
15      System.out.println("Marks 2: " + m2);
16      System.out.println("Marks 3: " + m3);
17      System.out.println("Total Marks: " + total);
18  }
19 }
```

OUTCOME-

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Marks 1: 85
Marks 2: 90
Marks 3: 78
Total Marks: 253
```

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 2 X
J AS4EX1.java > EmployeePayroll > main(String[] args)
1 import java.util.Scanner;
2
3 public class EmployeePayroll {
4     Run main | Debug main
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter Employee ID: ");
10        String idInput = sc.nextLine();
11
12        System.out.print("Enter Basic Salary: ");
13        String salaryInput = sc.nextLine();
14
15        System.out.print("Enter Bonus Amount: ");
16        String bonusInput = sc.nextLine();
17
18        Integer empId = Integer.valueOf(idInput);
19        Double basicSalary = Double.valueOf(salaryInput);
20        Double bonus = Double.valueOf(bonusInput);
21
22        if (basicSalary < 0 || bonus < 0) {
23            System.out.println("Invalid Salary or Bonus Amount!");
24        } else {
25            double netSalary = basicSalary + bonus;
26
27            System.out.println("\nEmployee ID: " + empId);
28            System.out.println("Basic Salary: " + basicSalary);
29            System.out.println("Bonus: " + bonus);
30            System.out.println("Net Salary: " + netSalary);
31        }
32
33        sc.close();
34    }
35 }
```

OUTCOME-

PROBLEMS 3

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Enter Employee ID:

→ 23

Enter Basic Salary:

→ 20000

Enter Bonus Amount:

→ 5000

Employee ID: 23

Basic Salary: 20000.0

Bonus: 5000.0

Net Salary: 25000.0